



# 12146 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

Cycle: 5, Proposal Category: GO

## INVESTIGATORS

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**OBSERVATIONS**

| <i>Folder</i> | <i>Observation</i> | <i>Label</i>   | <i>Observing Template</i>       | <i>Science Target</i> |
|---------------|--------------------|----------------|---------------------------------|-----------------------|
| Low-z-1       |                    |                |                                 |                       |
|               | 1                  | Low-z-1        | NIRSpec Fixed Slit Spectroscopy | (1) Low-z-1           |
|               | 2                  | Low-z-1_NRC_1  | NIRCam Imaging                  | (1) Low-z-1           |
|               | 3                  | Low-z-1_NRC_2  | NIRCam Imaging                  | (1) Low-z-1           |
| Low-z-2       |                    |                |                                 |                       |
|               | 5                  | Low-z-2        | NIRSpec Fixed Slit Spectroscopy | (2) Low-z-2           |
|               | 6                  | Low-z-2_NRC_1  | NIRCam Imaging                  | (2) Low-z-2           |
|               | 7                  | Low-z-2_NRC_2  | NIRCam Imaging                  | (2) Low-z-2           |
| Low-z-3       |                    |                |                                 |                       |
|               | 9                  | Low-z-3        | NIRSpec Fixed Slit Spectroscopy | (3) Low-z-3           |
|               | 10                 | Low-z-3_NRC_1  | NIRCam Imaging                  | (3) Low-z-3           |
|               | 11                 | Low-z-3_NRC_2  | NIRCam Imaging                  | (3) Low-z-3           |
| High-z-1      |                    |                |                                 |                       |
|               | 13                 | High-z-1       | NIRSpec Fixed Slit Spectroscopy | (5) High-z-1          |
|               | 14                 | High-z-1_NRC_1 | NIRCam Imaging                  | (5) High-z-1          |
|               | 15                 | High-z-1_NRC_2 | NIRCam Imaging                  | (5) High-z-1          |
| High-z-2      |                    |                |                                 |                       |
|               | 17                 | High-z-2       | NIRSpec Fixed Slit Spectroscopy | (6) High-z-2          |
|               | 18                 | High-z-2_NRC_1 | NIRCam Imaging                  | (6) High-z-2          |
|               | 19                 | High-z-2_NRC_2 | NIRCam Imaging                  | (6) High-z-2          |

**ABSTRACT**

The nature of dark energy is still one of the most confounding mysteries in astrophysics today, and the upcoming Roman Space Telescope will seek to shed light on its redshift evolution using Type Ia supernova (SN Ia) distances. That program will rely on SN Ia standardization at high-z being consistent with low-z, thus anomalous distance measurements in the early universe could indicate evolving dark energy and/or redshift-dependent SN Ia luminosity biases. Disentangling these effects is extremely difficult in the dark-energy-dominated universe ( $z < 1.5$ ), but by  $z > 2.5$ , dark energy is

vanishingly small and the Universe's expansion history is simple. SNe Ia should closely follow the expected distance-redshift relation in this range, and any deviations should not be caused by dark energy and thus would require differences in the SNe themselves. The first high- $z$  spectroscopic SN Ia sample from JWST ( $2 < z < 4$ ) indicates no such bias at  $z \sim 2$  but a possible trend at  $z > 3$ . This is beyond the reach of Roman to characterize, but would bias the Roman SN Ia cosmology measurement by 0.05 mag when propagated to  $z < 1.5$ . Here, we propose to add five  $z > 3$  SNe Ia to the high- $z$  sample with high S/N spectroscopic and photometric measurements that will both securely characterize any such luminosity bias, and enable the first detailed high- $z$  spectroscopic comparison studies to the low- $z$  sample. Spectroscopically confirming SNe Ia at  $z > 3$  will only be possible with JWST for the foreseeable future, giving each new addition to the sample enormous legacy value for future analyses.

## **OBSERVING DESCRIPTION**

This program contains 5 non-disruptive ToO triggers with carry-over status, each containing two visits. The first visit includes both nircam imaging and nirspec spectroscopy, the second nircam only. We will optimize the choices of filters, depths, and cadence depending on each discovery but have included fiducial follow-up strategies based on our expected redshift ranges and phases of discovery.

Proposal 12146 - Targets - Standard-ish Candles: Measuring SN Ia Luminosity Drift at  $z > 3$

| Generic Targets | #   | Name     | Criteria                         | Description |
|-----------------|-----|----------|----------------------------------|-------------|
|                 | (1) | Low-z-1  | $3 < z < 3.5$ SN Ia              |             |
|                 | (2) | Low-z-2  | $3 < z < 3.5$ SN Ia              |             |
|                 | (3) | Low-z-3  | $3 < z < 3.5$ SN Ia              |             |
|                 | (5) | High-z-1 | $z > 3.5$ SN Ia                  |             |
|                 | (6) | High-z-2 | $z > 3.5$ SN Ia                  |             |
|                 | (7) | OffsetTA | When an offset TA star is needed |             |
|                 |     |          |                                  |             |

# Proposal 12146 - Observation 1 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                                                     |                                                                                                               |                          |               |                 |            |                  |            |                  |                    |                     |                              |                 |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------|---------------|-----------------|------------|------------------|------------|------------------|--------------------|---------------------|------------------------------|-----------------|
| Observation                                         | Proposal 12146, Observation 1: Low-z-1                                                                        |                          |               |                 |            |                  |            |                  |                    |                     | Wed May 06 15:00:19 GMT 2026 |                 |
|                                                     | Diagnostic Status: Warning                                                                                    |                          |               |                 |            |                  |            |                  |                    |                     |                              |                 |
| Observing Template: NIRSpec Fixed Slit Spectroscopy |                                                                                                               |                          |               |                 |            |                  |            |                  |                    |                     |                              |                 |
| Diagnostics                                         | (Low-z-1 (Obs 1)) Warning (Form): Observers are responsible for checking that target acquisition is feasible. |                          |               |                 |            |                  |            |                  |                    |                     |                              |                 |
|                                                     | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                   |                          |               |                 |            |                  |            |                  |                    |                     |                              |                 |
| Generic Targets                                     | #                                                                                                             | Name                     | Criteria      |                 |            |                  |            |                  |                    |                     |                              | Description     |
|                                                     | (1)                                                                                                           | Low-z-1                  | 3<z<3.5 SN Ia |                 |            |                  |            |                  |                    |                     |                              |                 |
| Acquisition                                         | #                                                                                                             | Target                   | TA Method     | Subarray        | Filter     | Readout Pattern  | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID              |                 |
|                                                     | 1                                                                                                             | 7 OffsetTA               | WATA          | FULL            | F110W      | NRSRAPIDD6       | 3          | 1                | 1                  | 171.788             |                              |                 |
| Template                                            | HFF Readout Mode                                                                                              |                          |               |                 | Slit       |                  |            |                  | Subarray           |                     |                              |                 |
|                                                     | false                                                                                                         |                          |               |                 | S200A1     |                  |            |                  | FULL               |                     |                              |                 |
| Dithers                                             | #                                                                                                             | Primary Dither Positions |               |                 |            |                  |            |                  | Sub-Pixel Pattern  |                     |                              |                 |
|                                                     | 1                                                                                                             | 5                        |               |                 |            |                  |            |                  | NONE               |                     |                              |                 |
| Spectral Elements                                   | #                                                                                                             | Grating/Filter           | Slit          | Readout Pattern | Groups/Int | Integrations/Exp | #          | Autocal          | Total Dithers      | Total Integrations  | Total Exposure Time          | Optional ETC ID |
|                                                     | 1                                                                                                             | PRISM/CLEAR              | S200A1        | NRSIRS2         | 13         | 7                | 1          | NONE             | 5                  | 35                  | 33700.336                    |                 |

Proposal 12146 - Observation 1 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at  $z > 3$

|                      |                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | <div>On Hold ToO</div> <div>Target Of Opportunity Response Time 14 Days, Carry-Over</div> <div>Sequence Observations 1, 2, Non-interruptible</div> |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|

# Proposal 12146 - Observation 2 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                                 |                     |               |                 |            |                      |                    |               |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | <p>Proposal 12146, Observation 2: Low-z-1_NRC_1</p> <p>Diagnostic Status: Warning</p> <p>Observing Template: NIRCam Imaging</p> |                     |               |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                     |                     |               |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                               | Name                | Criteria      | Description     |            |                      |                    |               |                     |                              |
|                      | (1)                                                                                                                             | Low-z-1             | 3<z<3.5 SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                                          |                     |               |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                               |                     |               |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                               | Primary Dither Type |               | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                               | INTRAMODULEX        |               | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                               | Short Filter        | Long Filter   | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                               | F150W               | F356W         | SHALLOW4        | 8          | 1                    | 4                  | 4             | 1674.936            |                              |
|                      | 2                                                                                                                               | F200W               | F277W         | BRIGHT2         | 7          | 1                    | 4                  | 4             | 601.259             |                              |
| Special Requirements | <p>On Hold ToO</p> <p>3 After 2 by 40 Days to 60 Days</p> <p>Sequence Observations 1, 2, Non-interruptible</p>                  |                     |               |                 |            |                      |                    |               |                     |                              |

# Proposal 12146 - Observation 3 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                   |                     |                     |                 |            |                      |                    |               |                     |                              |
|----------------------|-------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | Proposal 12146, Observation 3: Low-z-1_NRC_2<br>Diagnostic Status: Warning<br>Observing Template: NIRCcam Imaging |                     |                     |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                       |                     |                     |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                 | Name                | Criteria            | Description     |            |                      |                    |               |                     |                              |
|                      | (1)                                                                                                               | Low-z-1             | $3 < z < 3.5$ SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                            |                     |                     |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                 |                     |                     |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                 | Primary Dither Type |                     | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                 | INTRAMODULEX        |                     | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                 | Short Filter        | Long Filter         | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                 | F150W               | F356W               | SHALLOW4        | 8          | 1                    | 4                  | 4             | 1674.936            |                              |
|                      | 2                                                                                                                 | F200W               | F277W               | SHALLOW4        | 6          | 1                    | 4                  | 4             | 1245.465            |                              |
| Special Requirements | On Hold Follow epoch 1 by 10-15 rest frame days.<br>3 After 2 by 40 Days to 60 Days                               |                     |                     |                 |            |                      |                    |               |                     |                              |



# Proposal 12146 - Observation 5 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                                                     |                                                                                                               |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------|---------------|-----------------|-------------|------------------|------------|------------------|--------------------|---------------------|------------------------------|-----------------|
| Observation                                         | Proposal 12146, Observation 5: Low-z-2                                                                        |                          |               |                 |             |                  |            |                  |                    |                     | Wed May 06 15:00:19 GMT 2026 |                 |
|                                                     | Diagnostic Status: Warning                                                                                    |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
| Observing Template: NIRSpec Fixed Slit Spectroscopy |                                                                                                               |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
| Diagnostics                                         | (Low-z-2 (Obs 5)) Warning (Form): Observers are responsible for checking that target acquisition is feasible. |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
|                                                     | (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                   |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
| Generic Targets                                     | #                                                                                                             | Name                     | Criteria      |                 | Description |                  |            |                  |                    |                     |                              |                 |
|                                                     | (2)                                                                                                           | Low-z-2                  | 3<z<3.5 SN Ia |                 |             |                  |            |                  |                    |                     |                              |                 |
| Acquisition                                         | #                                                                                                             | Target                   | TA Method     | Subarray        | Filter      | Readout Pattern  | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID              |                 |
|                                                     | 1                                                                                                             | 7 OffsetTA               | WATA          | FULL            | F110W       | NRSRAPIDD6       | 3          | 1                | 1                  | 171.788             |                              |                 |
| Template                                            | HFF Readout Mode                                                                                              |                          |               |                 | Slit        |                  |            |                  | Subarray           |                     |                              |                 |
|                                                     | false                                                                                                         |                          |               |                 | S200A1      |                  |            |                  | FULL               |                     |                              |                 |
| Dithers                                             | #                                                                                                             | Primary Dither Positions |               |                 |             |                  |            |                  | Sub-Pixel Pattern  |                     |                              |                 |
|                                                     | 1                                                                                                             | 5                        |               |                 |             |                  |            |                  | NONE               |                     |                              |                 |
| Spectral Elements                                   | #                                                                                                             | Grating/Filter           | Slit          | Readout Pattern | Groups/Int  | Integrations/Exp | #          | Autocal          | Total Dithers      | Total Integrations  | Total Exposure Time          | Optional ETC ID |
|                                                     | 1                                                                                                             | PRISM/CLEAR              | S200A1        | NRSIRS2         | 13          | 7                | 1          | NONE             | 5                  | 35                  | 33700.336                    |                 |

Proposal 12146 - Observation 5 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at  $z > 3$

|                      |                                                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | On Hold ToO<br>Target Of Opportunity Response Time 14 Days, Carry-Over<br>Sequence Observations 5, 6, Non-interruptible |
|----------------------|-------------------------------------------------------------------------------------------------------------------------|

# Proposal 12146 - Observation 6 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                                 |                     |               |                 |            |                      |                    |               |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | <p>Proposal 12146, Observation 6: Low-z-2_NRC_1</p> <p>Diagnostic Status: Warning</p> <p>Observing Template: NIRCam Imaging</p> |                     |               |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                     |                     |               |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                               | Name                | Criteria      | Description     |            |                      |                    |               |                     |                              |
|                      | (2)                                                                                                                             | Low-z-2             | 3<z<3.5 SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                                          |                     |               |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                               |                     |               |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                               | Primary Dither Type |               | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                               | INTRAMODULEX        |               | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                               | Short Filter        | Long Filter   | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                               | F150W               | F356W         | SHALLOW4        | 8          | 1                    | 4                  | 4             | 1674.936            |                              |
|                      | 2                                                                                                                               | F200W               | F277W         | BRIGHT2         | 7          | 1                    | 4                  | 4             | 601.259             |                              |
| Special Requirements | <p>On Hold ToO</p> <p>7 After 6 by 40 Days to 60 Days</p> <p>Sequence Observations 5, 6, Non-interruptible</p>                  |                     |               |                 |            |                      |                    |               |                     |                              |

# Proposal 12146 - Observation 7 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                   |                     |               |                 |            |                      |                    |               |                     |                              |
|----------------------|-------------------------------------------------------------------------------------------------------------------|---------------------|---------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | Proposal 12146, Observation 7: Low-z-2_NRC_2<br>Diagnostic Status: Warning<br>Observing Template: NIRCcam Imaging |                     |               |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                       |                     |               |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                 | Name                | Criteria      | Description     |            |                      |                    |               |                     |                              |
|                      | (2)                                                                                                               | Low-z-2             | 3<z<3.5 SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                            |                     |               |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                 |                     |               |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                 | Primary Dither Type |               | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                 | INTRAMODULEX        |               | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                 | Short Filter        | Long Filter   | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                 | F150W               | F356W         | SHALLOW4        | 8          | 1                    | 4                  | 4             | 1674.936            |                              |
|                      | 2                                                                                                                 | F200W               | F277W         | SHALLOW4        | 6          | 1                    | 4                  | 4             | 1245.465            |                              |
| Special Requirements | On Hold Follow epoch 1 by 10-15 rest frame days.<br>7 After 6 by 40 Days to 60 Days                               |                     |               |                 |            |                      |                    |               |                     |                              |

# Proposal 12146 - Observation 9 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                   |                                                                                                               |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
|-------------------|---------------------------------------------------------------------------------------------------------------|--------------------------|---------------|-----------------|-------------|------------------|------------|------------------|--------------------|---------------------|------------------------------|-----------------|
| Observation       | Proposal 12146, Observation 9: Low-z-3                                                                        |                          |               |                 |             |                  |            |                  |                    |                     | Wed May 06 15:00:19 GMT 2026 |                 |
|                   | Diagnostic Status: Warning                                                                                    |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
|                   | Observing Template: NIRSpec Fixed Slit Spectroscopy                                                           |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
| Diagnostics       | (Low-z-3 (Obs 9)) Warning (Form): Observers are responsible for checking that target acquisition is feasible. |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
|                   | (Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                   |                          |               |                 |             |                  |            |                  |                    |                     |                              |                 |
| Generic Targets   | #                                                                                                             | Name                     | Criteria      |                 | Description |                  |            |                  |                    |                     |                              |                 |
|                   | (3)                                                                                                           | Low-z-3                  | 3<z<3.5 SN Ia |                 |             |                  |            |                  |                    |                     |                              |                 |
| Acquisition       | #                                                                                                             | Target                   | TA Method     | Subarray        | Filter      | Readout Pattern  | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID              |                 |
|                   | 1                                                                                                             | 7 OffsetTA               | WATA          | FULL            | F110W       | NRSRAPIDD6       | 3          | 1                | 1                  | 171.788             |                              |                 |
| Template          | HFF Readout Mode                                                                                              |                          |               |                 | Slit        |                  |            |                  | Subarray           |                     |                              |                 |
|                   | false                                                                                                         |                          |               |                 | S200A1      |                  |            |                  | FULL               |                     |                              |                 |
| Dithers           | #                                                                                                             | Primary Dither Positions |               |                 |             |                  |            |                  | Sub-Pixel Pattern  |                     |                              |                 |
|                   | 1                                                                                                             | 5                        |               |                 |             |                  |            |                  | NONE               |                     |                              |                 |
| Spectral Elements | #                                                                                                             | Grating/Filter           | Slit          | Readout Pattern | Groups/Int  | Integrations/Exp | #          | Autocal          | Total Dithers      | Total Integrations  | Total Exposure Time          | Optional ETC ID |
|                   | 1                                                                                                             | PRISM/CLEAR              | S200A1        | NRSIRS2         | 13          | 7                | 1          | NONE             | 5                  | 35                  | 33700.336                    |                 |

Proposal 12146 - Observation 9 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at  $z > 3$

|                      |                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | On Hold ToO<br>Target Of Opportunity Response Time 14 Days, Carry-Over<br>Sequence Observations 9, 10, Non-interruptible |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|

# Proposal 12146 - Observation 10 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                   |                     |               |                 |            |                      |                    |               |                     |                              |
|----------------------|-------------------------------------------------------------------------------------------------------------------|---------------------|---------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | Proposal 12146, Observation 10: Low-z-3_NRC_1<br>Diagnostic Status: Warning<br>Observing Template: NIRCam Imaging |                     |               |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                      |                     |               |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                 | Name                | Criteria      | Description     |            |                      |                    |               |                     |                              |
|                      | (3)                                                                                                               | Low-z-3             | 3<z<3.5 SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                            |                     |               |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                 |                     |               |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                 | Primary Dither Type |               | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                 | INTRAMODULEX        |               | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                 | Short Filter        | Long Filter   | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                 | F150W               | F356W         | SHALLOW4        | 8          | 1                    | 4                  | 4             | 1674.936            |                              |
|                      | 2                                                                                                                 | F200W               | F277W         | BRIGHT2         | 7          | 1                    | 4                  | 4             | 601.259             |                              |
| Special Requirements | On Hold ToO<br>11 After 10 by 40 Days to 60 Days<br>Sequence Observations 9, 10, Non-interruptible                |                     |               |                 |            |                      |                    |               |                     |                              |

# Proposal 12146 - Observation 11 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                    |                     |               |                 |            |                      |                    |               |                     |                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------|---------------------|---------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | Proposal 12146, Observation 11: Low-z-3_NRC_2<br>Diagnostic Status: Warning<br>Observing Template: NIRCcam Imaging |                     |               |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                       |                     |               |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                  | Name                | Criteria      | Description     |            |                      |                    |               |                     |                              |
|                      | (3)                                                                                                                | Low-z-3             | 3<z<3.5 SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                             |                     |               |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                  |                     |               |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                  | Primary Dither Type |               | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                  | INTRAMODULEX        |               | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                  | Short Filter        | Long Filter   | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                  | F150W               | F356W         | SHALLOW4        | 8          | 1                    | 4                  | 4             | 1674.936            |                              |
|                      | 2                                                                                                                  | F200W               | F277W         | SHALLOW4        | 6          | 1                    | 4                  | 4             | 1245.465            |                              |
| Special Requirements | On Hold Follow epoch 1 by 10-15 rest frame days.<br>11 After 10 by 40 Days to 60 Days                              |                     |               |                 |            |                      |                    |               |                     |                              |



# Proposal 12146 - Observation 13 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                                                     |                                                                                                                 |                          |             |                 |             |                  |            |                  |                    |                     |                              |                 |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------|-------------|-----------------|-------------|------------------|------------|------------------|--------------------|---------------------|------------------------------|-----------------|
| Observation                                         | Proposal 12146, Observation 13: High-z-1                                                                        |                          |             |                 |             |                  |            |                  |                    |                     | Wed May 06 15:00:19 GMT 2026 |                 |
|                                                     | Diagnostic Status: Warning                                                                                      |                          |             |                 |             |                  |            |                  |                    |                     |                              |                 |
| Observing Template: NIRSpec Fixed Slit Spectroscopy |                                                                                                                 |                          |             |                 |             |                  |            |                  |                    |                     |                              |                 |
| Diagnostics                                         | (High-z-1 (Obs 13)) Warning (Form): Observers are responsible for checking that target acquisition is feasible. |                          |             |                 |             |                  |            |                  |                    |                     |                              |                 |
|                                                     | (Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                    |                          |             |                 |             |                  |            |                  |                    |                     |                              |                 |
|                                                     | (Visit 13:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                    |                          |             |                 |             |                  |            |                  |                    |                     |                              |                 |
| Generic Targets                                     | #                                                                                                               | Name                     | Criteria    |                 | Description |                  |            |                  |                    |                     |                              |                 |
|                                                     | (5)                                                                                                             | High-z-1                 | z>3.5 SN Ia |                 |             |                  |            |                  |                    |                     |                              |                 |
| Acquisition                                         | #                                                                                                               | Target                   | TA Method   | Subarray        | Filter      | Readout Pattern  | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID              |                 |
|                                                     | 1                                                                                                               | 7 OffsetTA               | WATA        | FULL            | F110W       | NRSRAPIDD6       | 3          | 1                | 1                  | 171.788             |                              |                 |
| Template                                            | HFF Readout Mode                                                                                                |                          |             |                 | Slit        |                  |            |                  | Subarray           |                     |                              |                 |
|                                                     | false                                                                                                           |                          |             |                 | S200A1      |                  |            |                  | FULL               |                     |                              |                 |
| Dithers                                             | #                                                                                                               | Primary Dither Positions |             |                 |             |                  |            |                  | Sub-Pixel Pattern  |                     |                              |                 |
|                                                     | 1                                                                                                               | 5                        |             |                 |             |                  |            |                  | SPATIAL            |                     |                              |                 |
| Spectral Elements                                   | #                                                                                                               | Grating/Filter           | Slit        | Readout Pattern | Groups/Int  | Integrations/Exp | #          | Autocal          | Total Dithers      | Total Integrations  | Total Exposure Time          | Optional ETC ID |
|                                                     | 1                                                                                                               | PRISM/CLEAR              | S200A1      | NRSIRS2         | 12          | 11               | 1          | NONE             | 10                 | 110                 | 97891.452                    |                 |

Proposal 12146 - Observation 13 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at  $z > 3$

|                      |                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | Group Visits within 53.0 Days<br>Visits Same PA<br>On Hold ToO<br>Target Of Opportunity Response Time 14 Days, Carry-Over<br><br>Sequence Observations 13, 14 within 4 Days |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Proposal 12146 - Observation 14 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                    |                     |                 |                 |            |                      |                    |               |                     |                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | Proposal 12146, Observation 14: High-z-1_NRC_1<br>Diagnostic Status: Warning<br>Observing Template: NIRCам Imaging |                     |                 |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                       |                     |                 |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                  | Name                | Criteria        | Description     |            |                      |                    |               |                     |                              |
|                      | (5)                                                                                                                | High-z-1            | $z > 3.5$ SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                             |                     |                 |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                  |                     |                 |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                  | Primary Dither Type |                 | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                  | INTRAMODULEX        |                 | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                  | Short Filter        | Long Filter     | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                  | F200W               | F356W           | SHALLOW4        | 5          | 1                    | 4                  | 4             | 1030.73             |                              |
|                      | 2                                                                                                                  | F200W               | F277W           | BRIGHT2         | 7          | 1                    | 4                  | 4             | 601.259             |                              |
| Special Requirements | On Hold ToO<br>15 After 14 by 50 Days to 100 Days<br>Sequence Observations 13, 14 within 4 Days                    |                     |                 |                 |            |                      |                    |               |                     |                              |

# Proposal 12146 - Observation 15 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                     |                     |                 |                 |            |                      |                    |               |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | Proposal 12146, Observation 15: High-z-1_NRC_2<br>Diagnostic Status: Warning<br>Observing Template: NIRCcam Imaging |                     |                 |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                        |                     |                 |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                   | Name                | Criteria        | Description     |            |                      |                    |               |                     |                              |
|                      | (5)                                                                                                                 | High-z-1            | $z > 3.5$ SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                              |                     |                 |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                   |                     |                 |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                   | Primary Dither Type |                 | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                   | INTRAMODULEX        |                 | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                   | Short Filter        | Long Filter     | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                   | F200W               | F356W           | SHALLOW4        | 10         | 2                    | 8                  | 4             | 4251.761            |                              |
|                      | 2                                                                                                                   | F200W               | F277W           | SHALLOW4        | 7          | 1                    | 4                  | 4             | 1460.201            |                              |
| Special Requirements | On Hold Follow epoch 1 by 10-15 rest frame days.<br>15 After 14 by 50 Days to 100 Days                              |                     |                 |                 |            |                      |                    |               |                     |                              |

# Proposal 12146 - Observation 17 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                   |                                                                                                                 |                          |             |                 |             |                  |            |                   |                    |                     |                              |                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------|-------------|-----------------|-------------|------------------|------------|-------------------|--------------------|---------------------|------------------------------|-----------------|
| Observation       | Proposal 12146, Observation 17: High-z-2                                                                        |                          |             |                 |             |                  |            |                   |                    |                     | Wed May 06 15:00:19 GMT 2026 |                 |
|                   | Diagnostic Status: Warning                                                                                      |                          |             |                 |             |                  |            |                   |                    |                     |                              |                 |
|                   | Observing Template: NIRSpec Fixed Slit Spectroscopy                                                             |                          |             |                 |             |                  |            |                   |                    |                     |                              |                 |
| Diagnostics       | (High-z-2 (Obs 17)) Warning (Form): Observers are responsible for checking that target acquisition is feasible. |                          |             |                 |             |                  |            |                   |                    |                     |                              |                 |
|                   | (Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                    |                          |             |                 |             |                  |            |                   |                    |                     |                              |                 |
|                   | (Visit 17:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                    |                          |             |                 |             |                  |            |                   |                    |                     |                              |                 |
| Generic Targets   | #                                                                                                               | Name                     | Criteria    |                 | Description |                  |            |                   |                    |                     |                              |                 |
|                   | (6)                                                                                                             | High-z-2                 | z>3.5 SN Ia |                 |             |                  |            |                   |                    |                     |                              |                 |
| Acquisition       | #                                                                                                               | Target                   | TA Method   | Subarray        | Filter      | Readout Pattern  | Groups/Int | Integrations/Exp  | Total Integrations | Total Exposure Time | Optional ETC ID              |                 |
|                   | 1                                                                                                               | 7 OffsetTA               | WATA        | FULL            | F110W       | NRSRAPIDD6       | 3          | 1                 | 1                  | 171.788             |                              |                 |
| Template          | HFF Readout Mode                                                                                                |                          |             |                 | Slit        |                  |            |                   | Subarray           |                     |                              |                 |
|                   | false                                                                                                           |                          |             |                 | S200A1      |                  |            |                   | FULL               |                     |                              |                 |
| Dithers           | #                                                                                                               | Primary Dither Positions |             |                 |             |                  |            | Sub-Pixel Pattern |                    |                     |                              |                 |
|                   | 1                                                                                                               | 5                        |             |                 |             |                  |            | SPATIAL           |                    |                     |                              |                 |
| Spectral Elements | #                                                                                                               | Grating/Filter           | Slit        | Readout Pattern | Groups/Int  | Integrations/Exp | #          | Autocal           | Total Dithers      | Total Integrations  | Total Exposure Time          | Optional ETC ID |
|                   | 1                                                                                                               | PRISM/CLEAR              | S200A1      | NRSIRS2         | 12          | 11               | 1          | NONE              | 10                 | 110                 | 97891.452                    |                 |

Proposal 12146 - Observation 17 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at  $z > 3$

|                      |                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | Group Visits within 53.0 Days<br>Visits Same PA<br>On Hold ToO<br>Target Of Opportunity Response Time 14 Days, Carry-Over<br><br>Sequence Observations 17, 18 within 4 Days |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Proposal 12146 - Observation 18 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                    |                     |                 |                 |            |                      |                    |               |                     |                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | Proposal 12146, Observation 18: High-z-2_NRC_1<br>Diagnostic Status: Warning<br>Observing Template: NIRCам Imaging |                     |                 |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                       |                     |                 |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                  | Name                | Criteria        | Description     |            |                      |                    |               |                     |                              |
|                      | (6)                                                                                                                | High-z-2            | $z > 3.5$ SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                             |                     |                 |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                  |                     |                 |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                  | Primary Dither Type |                 | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                  | INTRAMODULEX        |                 | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                  | Short Filter        | Long Filter     | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                  | F200W               | F356W           | SHALLOW4        | 5          | 1                    | 4                  | 4             | 1030.73             |                              |
|                      | 2                                                                                                                  | F200W               | F277W           | BRIGHT2         | 7          | 1                    | 4                  | 4             | 601.259             |                              |
| Special Requirements | On Hold ToO<br>19 After 18 by 50 Days to 100 Days<br>Sequence Observations 17, 18 within 4 Days                    |                     |                 |                 |            |                      |                    |               |                     |                              |

# Proposal 12146 - Observation 19 - Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$

|                      |                                                                                                                     |                     |                 |                 |            |                      |                    |               |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|-----------------|------------|----------------------|--------------------|---------------|---------------------|------------------------------|
| Observation          | Proposal 12146, Observation 19: High-z-2_NRC_2<br>Diagnostic Status: Warning<br>Observing Template: NIRCcam Imaging |                     |                 |                 |            |                      |                    |               |                     | Wed May 06 15:00:19 GMT 2026 |
| Diagnostics          | (Visit 19:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                        |                     |                 |                 |            |                      |                    |               |                     |                              |
| Generic Targets      | #                                                                                                                   | Name                | Criteria        | Description     |            |                      |                    |               |                     |                              |
|                      | (6)                                                                                                                 | High-z-2            | $z > 3.5$ SN Ia |                 |            |                      |                    |               |                     |                              |
| Template             | Module                                                                                                              |                     |                 |                 | Subarray   |                      |                    |               |                     |                              |
|                      | B                                                                                                                   |                     |                 |                 | FULL       |                      |                    |               |                     |                              |
| Dithers              | #                                                                                                                   | Primary Dither Type |                 | Primary Dithers |            | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions           |
|                      | 1                                                                                                                   | INTRAMODULEX        |                 | 4               |            | STANDARD             |                    |               |                     | 1                            |
| Spectral Elements    | #                                                                                                                   | Short Filter        | Long Filter     | Readout Pattern | Groups/Int | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                   | F200W               | F356W           | SHALLOW4        | 10         | 2                    | 8                  | 4             | 4251.761            |                              |
|                      | 2                                                                                                                   | F200W               | F277W           | SHALLOW4        | 7          | 1                    | 4                  | 4             | 1460.201            |                              |
| Special Requirements | On Hold Follow epoch 1 by 10-15 rest frame days.<br>19 After 18 by 50 Days to 100 Days                              |                     |                 |                 |            |                      |                    |               |                     |                              |